DEEPAK YADAV *Senior Data Scientist*

Pune, India dyyadav849@gmail.com +918427059627 Linkedin GitHub

Experienced Software Engineer and Data Scientist with expertise in building scalable, production-grade machine learning systems, specializing in search optimization and natural language processing (NLP). Demonstrated proficiency in deep learning, anomaly detection, and predictive maintenance, with a focus on integrating Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to enhance FAQ chatbots. Extensive experience in computer vision and Graph Neural Networks (GNNs). Skilled in leveraging GenAI and modern cloud infrastructure to deliver impactful, cutting-edge solutions.

SKILL

- **Languages**: Python, SQL, C++, OOPs with Data Structure
- **Machine Learning Frameworks**: PyTorch, TensorFlow, ONNX, PytorchLib, Langchain, Hugging Face
- **Cloud Platforms & Tools:** Azure ML, Databricks, MlFlow
- **MLOps & DevOps Tools**: Git, Gitlab, DVC, Flask, FastAPI, CI/CD, Docker, Kubernetes
- **ML Techniques:** Decision Trees, Random Forest, Bagging & Boosting, Clustering, Regularization, Anomaly Detection
- **Visualization & Processing:** Matplotlib, Seaborn, Pyspark, Pandas.
- **Domain Expertise**: NLP, Computer Vision, Transformer models, GNN

PROFESSIONAL EXPERIENCE

Senior Data Scientist at Bajaj Finserv 05/2023 – present
Pune, India

- <u>Intelligent FAQ and Query Resolution:</u> Developed an intelligent loan document query system that processes PDFs and web content to deliver accurate, context-aware responses. Implemented Retrieval-Augmented Generation (RAG) with vector-based retrieval using GPT and LLaMA for enhanced document understanding, achieving a 30% improvement in query resolution accuracy across multiple loan products.

- <u>Expense Manager Automation:</u> Built a scalable Expense Manager System for an NBFC using the OneMoney API to fetch and process millions of customer transactions securely. Automated key information extraction (merchant names, payment methods, transaction types) using Azure Databricks, PySpark, and a fine-tuned BERT model, reducing manual effort by 30% and improving extraction accuracy by 22%.

- <u>Address Standardization Pipeline:</u> Developed pipeline using Databricks and PySpark to process millions of records, applying Fuzzy Matching (Levenshtein, Jaro-Winkler) for deduplication and error correction. Integrated OpenStreetMap API for geolocation to fill missing address details and used BERT-based NLP models for contextual entity recognition, improving accuracy in parsing address components like street names and postal codes. Automated with Apache Airflow and tracked metrics with MLflow, achieving 99% standardization accuracy.

Machine Learning Engineer at SONY 12/2022 – 05/2023
Bangalore, India

- Developed a real-time pose detection system using STGCN trained in PyTorch and deployed with a C++ API via Crow. Optimized the model using TorchScript, quantization, and model pruning for efficient edge device inference in applications like fitness tracking and surveillance.

Data Scientist at Stealth Start-up 05/2022 – 12/2022
Bangalore, India

- We developed a drift detection system using VAE and GAN to handle object detection failures in foggy and low-light conditions. We implemented a reconstruction pipeline to convert video frames from foggy/dark to well-lit conditions, maintaining detection accuracy. Achieved a 30% boost in object detection using A/B testing for validation

Machine Learning Research Engineer at INRIA, Paris 09/2020 – 05/2022
Paris, France

Project - 1 :

- Worked on multilingual datasets and built automated text preprocessing pipelines. Developed LLM-based pipelines to detect radicalization sentences on social media.

Project - 2 :

- Developed a multimodal machine learning pipeline for emotion recognition from video, EEG, ECG, EDA signals, and transcript-text data. Achieved high accuracy in classifying human emotions on arousal and valence scales.

Summer Research Intern at National Taiwan University 06/2020 – 08/2020
Hualien, Taiwan

- Worked on Project for Sq2Sq Machine Translation of Indigenous Indian Languages(Bengali, Tamil, and Telugu). I've used attention-based LSTM cell-based RNN, encoder & decoder architecture to generate indic language. We got the average BLUE score of 4.07, 5.43, and 4.97, respectively, languages trained on 25000 large text corpora.

EDUCATION

Master in Machine Learning and AI at IIIT Bangalore, Bangalore, India
International Institute of Technology - IIT Bangalore

Bachelor of Engineering(B.E) in Computer Science & Engineering at CU, Chandigarh, India
Chandigarh University

Major Courses Work - Linear Algebra, Calculus and Vector Space, Discrete Mathematics, Probability & Statistics, Game and Graph Theory, Artificial Intelligence & Machine Learning, Data Structure & Analysis of Algorithms, Object-Oriented Programming,

PUBLICATION

- A. Bahuguna, D. Yadav, A. Senapati, B. N. Saha. **KNN-SVM with Deep Features for COVID-19 Pneumonia Detection from Chest-Xray.** (Accepted for Conference at International Conference on Mathematics and Computing (ICMC) 2022) **[Oral Presentation]**

- A. Bahuguna, D. Yadav, A. Senapati, B. N. Saha. **A Unified Deep Neuro-Fuzzy Approach for COVID-19 Twitter Sentiment Classification.** (Accepted for Journal at Journal of Intelligent & Fuzzy Systems (JIFS) 2022) **[Oral Presentation]**

AWARDS

- Selected for the **2020 MOST GASE Global Talent Summer Internship Program(April 2020) (Lab: Artificial Intelligence) organized by the Ministry of Science & Technology** and NTU. Selected as one of the 35 students from all over the world.
- Selected for the prestigious **Charpak Exchange (BCS) Scholarship: Awarded by the France Embassy** in India.
- Selected for the **Google Research Week Program(NLU Track**) organized by the Google Research team in India

PROJECTS

Semantic Segmentation Using U-Net for Satellite Imagery

Developed a U-Net model to segment satellite images into categories such as water, land, roads, buildings, and vegetation. Implemented an end-to-end pipeline with data preprocessing, augmentation, and real-time inference using a Streamlit application. The model is designed for applications like urban planning and environmental monitoring, offering pixel-level accuracy for large-scale land segmentation tasks.

- **Technologies: Python, TensorFlow/Keras, U-Net, OpenCV, Streamlit**

Cartoon Image Generator Using VIT And GPT Based Models

Developed a model combining Vision Transformers (ViT) and Generative Pre-trained Transformers (GPT) to convert images into cartoon-style outputs. Trained on a large dataset of image pairs and deployed as a web application.

Skills: Python, PyTorch, TensorFlow, Transformers (Hugging Face), OpenCV, Flask

HIV Classification using Graph Neural Network with Attention

Worked on Graph Convolutional Neural network with attention to classifying HIV using molecule-net data sets. Accuracy: 96.7%, Precision: 38.1% Recall: 43.9%, F1-Score: 40.8. Used GVAE to generate the graph-based molecules

Skill: **Pytorch Geometric, Graph Neural Network(GAT), GVAE, Deep Learning.**

Graph Neural Network-based Recommendation System

Developed a personalized product recommendation system using Graph Neural Networks (GNN) with the LightGCN model, designed for large-scale recommendation tasks. Built an end-to-end solution with a FastAPI backend for serving recommendations, and an HTML/JavaScript frontend for user interaction

- **Skills: LightGCN, PyTorch, FastAPI, GNN, HTML, JavaScript**

Fake News Detection Using Mutual Embedding via Graph Neural Network

Worked on social fake news datasets, Extracted news context level features using W2Vec and BERT then merged them with GNN Layers to get social context extraction. Got a TestF1 score of 92%.

Skills: GNN, W2Vec, BERT.

Graph-Based Credit Card Fraud Detection

- Developed a graph-based approach to detect fraudulent transactions in financial data, significantly improving accuracy and recall. Constructed a graph using customer and transaction data, and trained a Graph Neural Network (GNN) to identify patterns indicative of fraud. Deployed the model for real-time fraud detection, ensuring seamless integration with existing financial systems.

Skills: GNN, PyTorch Geometric, NetworkX, FastAPI, Docker.